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Comparative assessment of geostatistical and deterministic interpolation methods for seasonal groundwater level mapping in the Arpa River Basin, Central India

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Abstract

Groundwater level mapping is essential for groundwater resource assessment and sustainable water management. This study examines the performance of deterministic interpolation methodologies and geostatistical methods in mapping groundwater levels by analyzing the groundwater level data from 69 wells belonging to the Central Ground Water Board (CGWB) and covering the period from 2001 to 2025 in the Arpa River Basin in Central India. The seasonal datasets were arranged for the rainy season (June to September), winter season (October to January), and summer season (February to May) and were analyzed through different methods like Inverse Distance Weighting (IDW), Regularized Spline, and Ordinary Kriging with Circular, Spherical, Exponential and Gaussian semivariogram in ArcGIS. The model performance was analyzed through different cross-validation statistics like Mean Error (ME), Root Mean Square Error (RMSE), Mean Standardized Error (MSE), Root Mean Square Standardized Error (RMSSE), and Average Standard Error (ASE). The results indicate that geostatistical methods generally produced more reliable and spatially consistent groundwater level surfaces than deterministic approaches. In spite of the fact that the Exponential semivariogram produced lower prediction errors, the uncertainty-related parameters, in contrast, varied in the Circular, Spherical, and Gaussian models depending on the seasons. Therefore, no single semivariogram model was universally superior across all validation criteria, and model selection should be based on the combined interpretation of prediction accuracy and uncertainty measures. The seasonal groundwater level maps created during the research can serve as the scientific background for more efficient groundwater monitoring, as well as basin and groundwater management of the Arpa River Basin.

Keywords Groundwater table, Spatial interpolation, Ordinary kriging, Semivariogram analysis, Seasonal groundwater variability



1 Introduction

The use of groundwater as a source of fresh water for domestic, agricultural and industrial uses is an important freshwater resource for all communities, but most importantly within river basins in semi-arid areas like Central India [9, 39, 44]. Seasonal changes to groundwater levels due to factors like wet/dry season recharge, amounts of water pumped, differences in rainfall between seasons, and the physical property or hydrogeology (hydraulic) also affect how much water will be available at that location. To maintain sustainable use and management of groundwater, it is necessary to understand the distribution, both spatially and temporally, of groundwater levels for that location [19, 27, 36, 46]. However, in most cases, groundwater level data is available from just the wells that were sampled. Therefore, in order to obtain groundwater levels located between these wells it is necessary to map groundwater levels, using spatial interpolation techniques, on a GIS (Geographic Information System) to create continuous surfaces that represent the spatial behavior of groundwater within the entire basin [4, 18, 21, 23, 42, 45].

Interpolation techniques can be categorized into two types: Deterministic and Geostatistical. Deterministic interpolators (e.g., inverse distance weighting (IDW), spline) predict unknown values from mathematical relationships (or equations) that relate distance and spatial arrangement to observed points, without any reference to autocorrelation in the spatial network [32, 49]. Geostatistical techniques, on the other hand, usually employ Ordinary Kriging, Circular, Spherical, Exponential and Gaussian to produce predictions about variances and means on the basis of variogram modelling. These methods additionally allow the quantifying of uncertainty in the estimates through the use of statistics (e.g., confidence intervals). The selection of an appropriate interpolation method depends on data distribution, spatial variability, stationarity, and the underlying hydrogeological processes influencing groundwater levels [5, 8, 9, 24, 47, 50].

Spatial interpolation has been widely utilized in hydrology, environmental studies, and groundwater studies in the last few years for: resource assessment; vulnerability mapping; risk zonation; decision support based on multi-criteria approaches. As such, the averaged groundwater levels, average rainfall amounts, and other hydro-environmental variables are important input layers in a multi-criteria decision-making process ([5], [5]; [6, 17, 26, 28]). Examples of multi-criteria decision-making processes include: weighted overlay analysis; groundwater potential mapping; delineation of recharge areas; and sustainability assessments [11, 15, 24, 31]. The accuracy of the combined data analysis based on interpolation is very important because errors introduced during interpolation may propagate and greatly influence the final results. Consequently, an appropriate interpolation method must be chosen that accurately represents the spatial variability of the data [2]. Accurate interpolation can significantly enhance the realism of the spatial analysis, and increase the reliability of other analyses performed using the technique of weighted overlay in the evaluation of groundwater resources, recommendation of sustainable water use, and the establishment of planning initiatives for managing groundwater resources [22, 30, 40, 43].

The use of deterministic and geostatistical interpolation methods has become renowned in the mapping of groundwater levels as new studies reveal that while applicable across various environments, both of these methods effectiveness fluctuates depending on the specific hydrogeological conditions of the area, the manner in which spatial data is collected and the interpolation model that is being used. Although ordinary

kriging provides an acceptable method for space forecasting, and is able to model spatial autocorrelation thanks to semivariogram modeling, deterministic techniques such as inverse distance weighting (IDW) and spline would primarily depend on the spatial proximity of the points. Based on this evidence, the choice of a proper interpolation method should rely on the characteristics of the given region and not on the general applicability of the methods [22, 33, 37, 38].

Although several studies have compared groundwater interpolation methods, most have focused on annual groundwater conditions or evaluated a limited number of interpolation techniques [33, 34, 48, 51]. Relatively few studies have systematically compared deterministic and geostatistical methods for seasonal spatial prediction of groundwater levels using long-term monitoring data in Central Indian River basins. Consequently, the most suitable interpolation technique for generating reliable seasonal groundwater level surfaces remains uncertain for the Arpa River Basin. Previous comparative studies were generally limited to a single season, a single kriging model, or short-term datasets. In contrast, the present study evaluates four semivariogram models separately for three hydrological seasons using a long-term groundwater monitoring dataset and a common cross-validation framework [12, 20, 29].

In order to fill this gap, this study compares deterministic (IDW and Spline) and geostatistical (Ordinary Kriging) interpolation methods for seasonal estimation of groundwater levels at the Arpa River Basin based on groundwater monitoring data collected from 2001 to 2025. Four semivariogram models (Circular, Spherical, Exponential and Gaussian) were tested separately for rainy, winter and summer seasons to find the most suitable interpolation model for different groundwater situations [3, 13, 24, 35, 38]. Groundwater abstraction, recharge, and consumption are seasonally variable, which means that the spatial distribution of groundwater levels and the performance of the interpolation methods differ as well. Hence, the season specific model validation has been performed to select the best method in terms of groundwater level estimation using cross-validation statistics (ME, RMSE, ASE) [1, 14, 20, 31, 33, 38, 41]. The results of this study can have practical implications for the assessment and management of groundwater resources at the any watershed planning like Arpa River Basin.

2 Methodology

A study comparing the use of deterministic interpolations versus geostatistical methods for mapping groundwater level seasonality across the Arpa River Basin of Central India has been completed. Data were obtained from observation wells within the basin between 2001 and 2025 with respect to how deep the groundwater is located (i.e., depth below surface level). Groundwater surfaces composed of the geographical location of the groundwater at different times of the year were created using the Geostatistical Analyst and ArcGIS software applications [7, 10, 11, 16, 25]. Three types of interpolation methods were compared through statistical methods (i.e. the error metrics of Root Mean Square Error, Mean Error, etc.). By comparing these indications as well as the accuracy of all interpolation techniques and their spatial representation accuracy on the map, the researchers have been able to ascertain which interpolation technique is optimally suited for use in developing groundwater maps for the Arpa River Basin, thereby creating a scientifically sound basis for subsequent groundwater assessments and decision-making. Figure 1 shows the study area map of Arpa Basin with observation well.

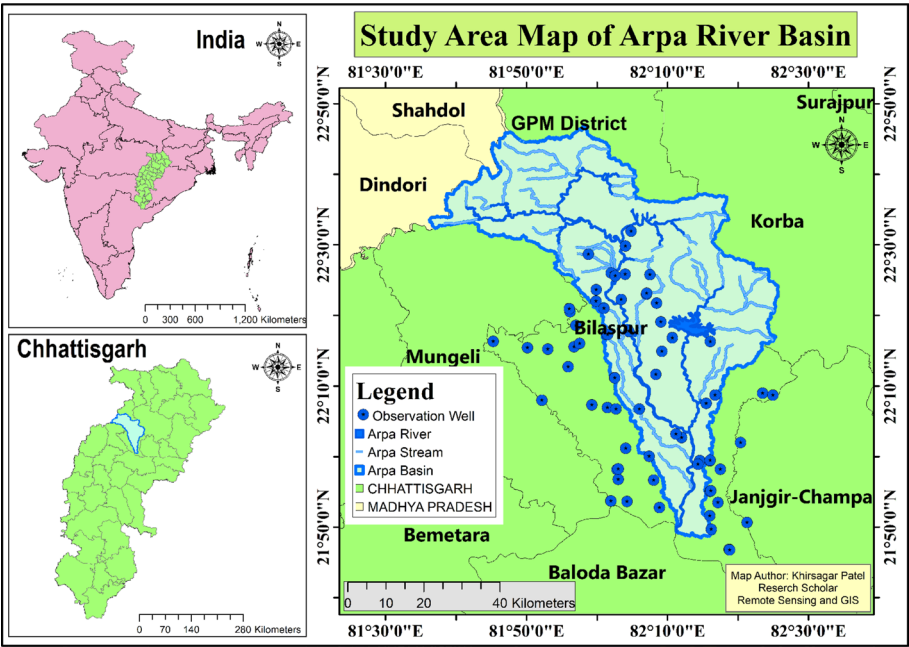


Fig. 1 Study area Arpa River Basin map with observation well

Table 1 Data source and informetric details of observation wells

Parameter	Description
Data Source	CGWB (North Central Chhattisgarh Region)
Study Period	2001–2025 (Seasonal- Rainy, Winter, Summer)
No. of Wells	69 Observation Wells
Parameter	Depth to Water Level
Unit	mbgl (meter below ground level)

2.1 Groundwater database

The groundwater level data used in this study were obtained from the pre-established groundwater observation well network maintained by the Central Ground Water Board (CGWB), Government of India, within the Arpa River Basin for the period 2001–2025. The collected groundwater level information refers to the distance from the water table to the ground level, measured in meters (m bgl). Thus, extensive data cleaning was carried out to determine whether the data met quality criteria such as completeness, duplication, absence of data, and spatial validity. After this, only valid records have remained for interpolation purposes. In addition, the observation wells were appropriately distributed in space to reflect groundwater connectivity in various hydrogeological environments (Table 1).

2.2 Data preparation

The groundwater observations were grouped into three hydrological seasons, namely Rainy (June–September), Winter (October–January), and Summer (February–May). Seasonal datasets were prepared by averaging groundwater level observations corresponding to each season. Prior to interpolation, the datasets were examined for outliers and consistency to minimize errors during spatial prediction. All groundwater observation well locations were projected into the WGS 1984 / UTM Zone 44 N coordinate

reference system to ensure spatial consistency during interpolation and map generation. Spatial interpolation was performed using an output raster cell size (spatial resolution) of 30 m, providing a uniform spatial framework for generating seasonal groundwater level maps across the Arpa River Basin.

The data concerning the groundwater levels were received from Central Ground Water Board (CGWB) and contained readings taken in 69 fixed groundwater monitoring wells in Arpa River Basin during time period 2001–2025. The monitoring network depicts the shallow aquifer conditions including dug and tube wells. Monthly groundwater measurements were classified into the three hydrological seasons, and for each observation well seasonal average values were computed in order to create georeferenced seasonal datasets. The dataset was analyzed for completeness, duplicate observations, etc., and no major missing observations were revealed in the dataset which would need imputation of data. The seasonal datasets developed were used for spatial interpolation and cross-validation in order to search for the most suitable interpolation method for groundwater resource studies, conservation, and watershed management.

In Fig. 2 shows the methodology flowchart of the present study, illustrating the sequential steps followed for seasonal groundwater level mapping in the Arpa River Basin. It includes data collection, preprocessing, interpolation using deterministic and geostatistical methods, validation, and comparative analysis to identify the most suitable technique.

2.3 Interpolation techniques

Spatial interpolation techniques are employed to estimate groundwater levels at unsampled locations using observations collected from monitoring wells. In the present study, deterministic and geostatistical interpolation techniques available in ArcGIS were

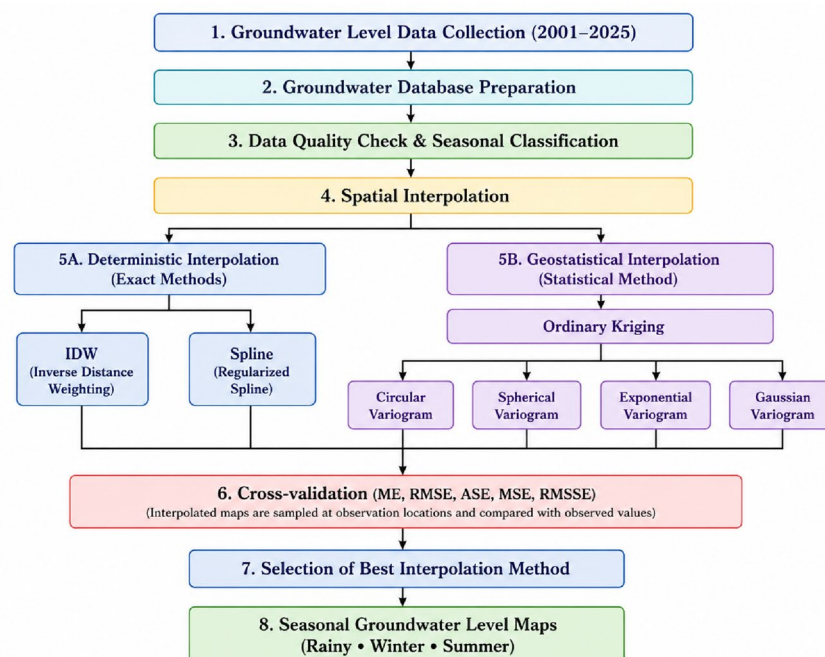


Fig. 2 Methodology flow chart of this study

applied to evaluate their capability in representing seasonal groundwater level variations within the Arpa River Basin.

2.3.1 Geostatistical Interpolation Method

2.3.1.1 Ordinary Kriging (OK) Ordinary Kriging (OK) is a geostatistical interpolation technique that estimates groundwater levels at unsampled locations by considering the spatial autocorrelation among observed data points. The method assumes that the local mean is unknown but constant within the search neighborhood and determines optimal weights using the fitted semivariogram model.

Kriging estimator:

$$Z(x_0) = \sum_{i=1}^n \lambda_i Z(x_i) \quad (1)$$

where, $Z(x_0)$ = predicted groundwater level.

λ_i = kriging weights.

$Z(x_i)$ = observed groundwater level.

n = number of observations.

Sum of weights = 1.

2.3.1.2 Semivariogram analysis The Semivariogram quantifies spatial dependence by relating variance to separation distance.

Experimental Semivariogram The experimental semivariogram quantifies the spatial dependence between groundwater observations by relating semivariance to the separation distance between observation pairs. It forms the basis for fitting theoretical semivariogram models used in Ordinary Kriging.

Formula:

$$\gamma(h) = \frac{1}{2N(h)} \sum_{i=1}^{N(h)} [Z(x_i) - Z(x_i + h)]^2 \quad (2)$$

where,

$\gamma(h)$ = semivariance

h = lag distance

$N(h)$ = number of points

$Z(x_i)$ = groundwater level.

2.3.1.3 Semivariogram parameters Nugget (C_0): Measurement error or micro-scale variation at zero distance.

Sill ($C_0 + C$): Total variance where Semivariogram levels off.

Range (a): Distance beyond which spatial correlation becomes negligible.

2.3.1.4 Semivariogram models used in ordinary kriging The experimental semivariogram was fitted using Circular, Spherical, Exponential and Gaussian theoretical models.

2.3.2 Circular model

The Circular semivariogram model describes a gradual increase in semivariance with distance until the sill is reached. This model is appropriate for datasets exhibiting moderate spatial continuity.

Formula:

$$\gamma(h) = \begin{cases} C_0 + C \left[1 - \frac{2}{\pi} \cos^{-1} \left(\frac{h}{a} \right) + \frac{2h}{\pi a} \sqrt{1 - \left(\frac{h}{a} \right)^2} \right], & h \leq a \\ C_0 + C, & h > a \end{cases} \quad (3)$$

Suitable where spatial continuity abruptly changes.

2.3.3 Spherical model

The Spherical semivariogram model is one of the most widely used models in hydrogeological studies because it represents spatial dependence that increases with distance and stabilizes at the sill after reaching the range.

Formula:

$$\gamma(h) = \begin{cases} C_0 + C \left[\frac{3h}{2a} - \frac{1}{2} \left(\frac{h}{a} \right)^3 \right], & h \leq a \\ C_0 + C, & h > a \end{cases} \quad (4)$$

Most commonly used in groundwater studies.

2.3.4 Exponential model

The Exponential semivariogram model represents spatial continuity that increases rapidly at short distances and gradually approaches the sill asymptotically. It is suitable for groundwater datasets exhibiting short-range spatial variability.

Formula:

$$\gamma(h) = C_0 + C \left[1 - \exp \left(-\frac{h}{a} \right) \right] \quad (5)$$

Spatial correlation decreases gradually; sill approached asymptotically.

2.3.5 Gaussian model

The Gaussian semivariogram model produces a very smooth spatial surface because semivariance increases gradually near the origin before approaching the sill. This model is often suitable for highly continuous groundwater datasets.

Formula:

$$\gamma(h) = C_0 + C \left[1 - \exp \left(-\frac{h^2}{a^2} \right) \right] \quad (6)$$

Represents very smooth spatial variation.

2.4 Deterministic interpolation methods

2.4.1 Inverse distance weighted (IDW)

IDW is a deterministic interpolation method that estimates unknown values based on the principle that nearby observations have a greater influence than distant ones. The

influence of each observation decreases with increasing distance from the prediction location.

Formula:

$$Z(x_0) = \frac{\sum_{i=1}^n Z(x_i) d_i^{-p}}{\sum_{i=1}^n d_i^{-p}} \quad (7)$$

where,

$Z(x_0)$ = interpolated value at unknown location

$Z(x_i)$ = observed value at known location i

d_i = distance between x_0 and x_i

p = power parameter controlling distance decay

n = number of surrounding points

Higher gives more weight to nearer points.

2.5 Spline Interpolation

Spline interpolation fits a smooth mathematical surface that passes exactly through the measured points and minimizes overall surface curvature. Regularized spline was adopted for groundwater interpolation because it generates smooth surfaces while minimizing overall surface curvature.

Formula:

$$Z(x, y) = T(x, y) + \sum_{i=1}^n \lambda_i R(r_i) \quad (8)$$

where,

$T(x, y)$ = trend function

λ_i = weight coefficients

$R(r_i)$ = radial basis function

(r_i) = distance from known point

n = number of points

Suitable for smooth surfaces but may overshoot in practical applications.

The objective of the present study was not to analyze long-term temporal trends in groundwater levels, but to generate representative long-term seasonal groundwater level surfaces for comparing the performance of deterministic and geostatistical interpolation methods. The cross-validation error analysis of the interpolation techniques and Semivariogram models used in this study was performed using five different error statistic measurements, Mean Error (ME), Root Mean Square Error (RMSE), Average Standard Error (ASE), Mean Standardized Error (MSE), and Root Mean Square Standardized error (RMSSE). The purpose of these statistics is to estimate the accuracy, bias, and uncertainty of the prediction methods, and to identify the best interpolation

method for mapping groundwater levels on a seasonal basis. Model performance was evaluated using the cross-validation tool available in ArcGIS. Cross-validation compares observed groundwater levels with predicted values obtained by sequentially removing one observation at a time and estimating its value from the remaining observations. The best interpolation method was selected based on standard cross-validation criteria: $ME \approx 0$, $MSE \approx 0$, minimum RMSE, $RMSE \approx ASE$, and $RMSSE \approx 1$. These statistics collectively indicate low prediction bias, high accuracy, reliable uncertainty estimation, and appropriate model performance.

In order to enable uniform visualization, the seasonal groundwater level maps were created in a standard method relying on ten different classes and a common colour gradient. Each of the interpolation techniques produced different ranges of predictions as a function of its mathematical properties, but slight differences in class break values were acceptable so that the original distribution of data was taken into account without distorting the interpolation process. Therefore, visual interpretation of the maps has been made in conjunction with the cross-validation data rather than based solely on the map looks.

3 Result and discussion

The findings from the application of spatial interpolation to groundwater level data are presented in this section using a variety of methods and models. An assessment was performed on the applicability and effectiveness of each method to determine which was the best option for seasonal groundwater table mapping in the study area.

3.1 Spatial interpolation framework for seasonal groundwater table mapping

Groundwater table levels are significantly variable seasonally and spatially as a consequence of their many controlling factors, including Aquifer Type, Precipitation events, Texture of the soil, Topography, and groundwater pumping rate with tube wells and Anthropogenic activities etc. Due to the high degree of variable aquifer characteristics across both space and time, a single method of interpolation will result in a large amount of spatial and temporal error; therefore, it is necessary to evaluate multiple methods of interpolation to find the best and most accurate methods for mapping the groundwater table on a seasonal basis.

3.2 Experimental semivariogram analysis average water table (mbgl) of 2001–2025 in rainy season

The experimental variogram shows a gradual increase in semivariance with distance, indicating strong spatial dependence of groundwater levels. A low nugget effect suggests minimal measurement error, while the variogram reaches a sill at approximately 20–35 km, representing the spatial range of groundwater level correlation in the basin.

Figure 3 shows the experimental Semivariogram and theoretical models (Circular, Spherical, Exponential, and Gaussian) that fit the experimental for average groundwater tables (mbgl) during the rainy season (2001–2025). The experimental Semivariogram indicates that as distance (lag distance) increases, semivariance increases, suggesting a strong spatial autocorrelation of groundwater levels throughout the basin. Also, there is evidence of moderate size nugget effects, indicating there is some measurement error and small-scale heterogeneity (i.e., where the variability of groundwater level is related to

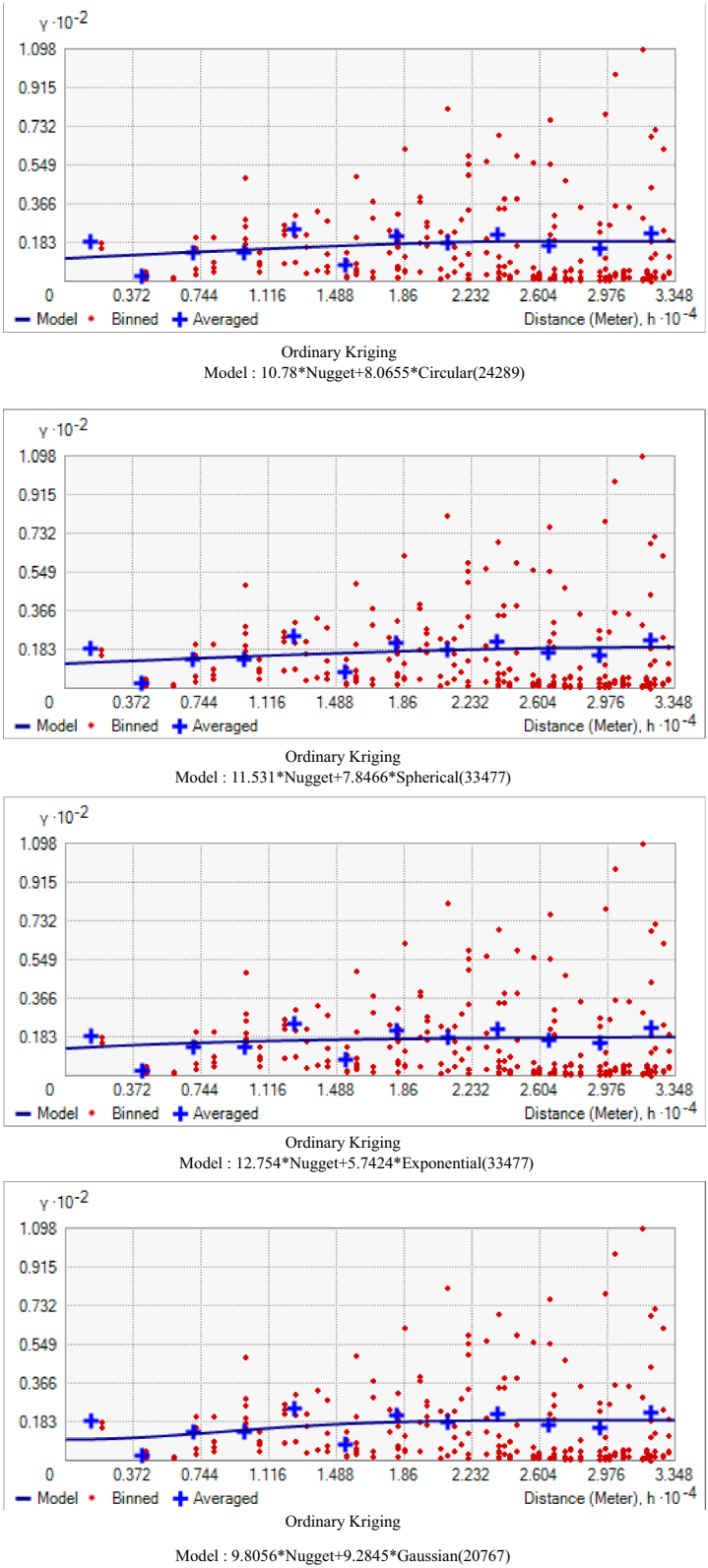


Fig. 3 Experimental Semivariogram and fitted circular model for average groundwater table (mbgl) during the rainy season

the heterogeneity/hydrogeology type (and/or mixed) of local wells). A sill, a flat segment in all of the models, provides evidence that groundwater level is variable only within a defined distance. This indicates that while groundwater levels may vary on a small (local) scale, the spatially structured spatial processes rather than those resulting from chance will be defined over a larger area. The theoretical ranges indicate that groundwater levels remain spatially correlated over basin-scale distances (approximately 20–33 km). The overall characteristics of the Semivariogram support a moderate to strong level of dependence of groundwater levels on spatial relationships at the time of the rainy season. Thus, Ordinary Kriging is an appropriate method for mapping seasonal groundwater tables in the Arpa River Basin.

The Circular model is the noticeable best method for modelling how groundwater levels in the Arpa River Basin connect to each other at short distances. The Circular model has lots of misalignment between actual measurements and theoretical predictions at longer distances. The Exponential model displays a gradual connection between groundwater levels across space and shows some correlation at long distances but underestimates the amount of groundwater level change over short distances which causes too much smoothing of the way groundwater levels change over time. The Gaussian model assumes that there is a very smooth connection between groundwater levels across space, but it does not properly describe the discontinuities that occur with semi variances at short lag distances which makes it not nearly as good for describing how groundwater levels vary during the rainy season. The Spherical model has the best correlation with measured semi variances across distances with a clear and well-defined sill and range making it the most accurate portrayal of both short and long-distance connections of groundwater levels. Therefore, the best method for mapping seasonal changes in the groundwater table (mbgl) in the Arpa River Basin during the rainy season is through Ordinary Kriging with the Spherical Semivariogram model.

3.2.1 Experimental Semivariogram analysis average water table (mbgl) of 2001–2025 in winter season

The experimental Semivariogram, calculated for the 2001–2025 winter season, exhibits in Fig. 4, a continual increase of semivariance values with increased lag distances. The strongest levels of spatial dependence can therefore be accomplished through moderate to strong relationships between groundwater levels. When compared with the experimental data results, the fitted Spherical Semivariogram provided the best description of basin-scale spatial continuity, as its sill and range were well defined in relation to the experimental data points.

The Circular model showed some accuracy in capturing short range variability for groundwater depths, while Exponential and Gaussian models tended to produce similar smooth curves for the entire range of lag distances but were not successful in capturing short lag distances' local variabilities. Thus, Ordinary Kriging utilizing the Spherical Semivariogram model will provide the best method for accurately mapping the winter season groundwater table (meter below ground level) across the Arpa River Basin. For Semivariogram parameters obtained, the Spherical and Exponential models have the longest range of about 33.5 km, signifying a good groundwater correlation between different locations in the entire basin, and the Circular model has a moderate effective

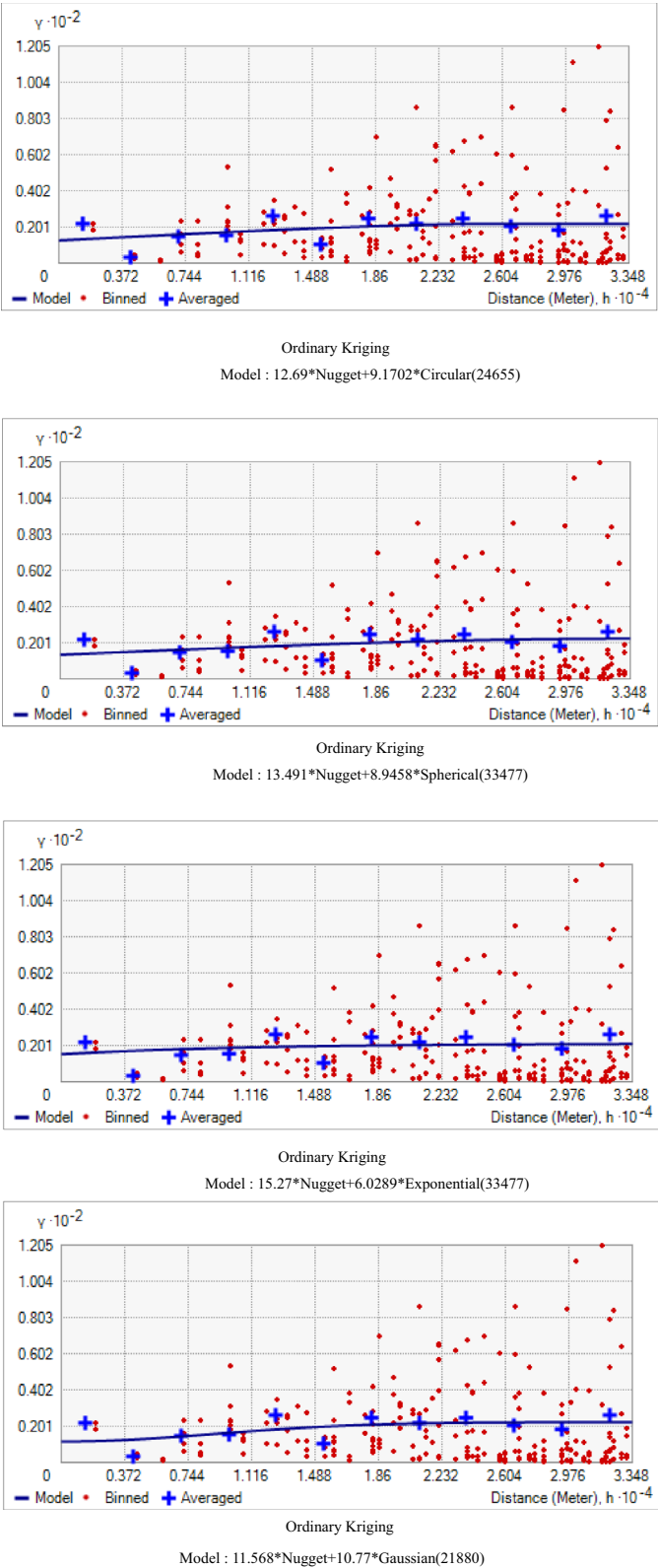


Fig. 4 Experimental semivariogram and fitted circular model for average groundwater table (mbgl) during the winter season

range (about 24.7 km), while the Gaussian model has the shortest range (about 21.9 km), which indicates greater smoothing and less accuracy at short distances between points.

3.3 Experimental Semivariogram analysis average water table (mbgl) of 2001–2025 in summer season

The experimental Semivariogram for Summer (2001–2025), indicates in Fig. 5, a weaker but still statistically significant spatial dependence between multiple measurements of groundwater depth, likely related to greater stress upon the aquifer from reduced recharge and higher rates of abstraction. While the Spherical model provides the best overall fit with the observed experimental Semivariogram by accurately representing both short- and medium-distance variability with stable sill and range, the Circular model is an adequate representation of short-distance continuity; however, both the Exponential and Gaussian models produce excessive smoothness for localized variations in groundwater depth readings. Based on these observations, Ordinary Kriging with the Spherical Semivariogram will provide the best method for mapping the groundwater table (mbgl) for summertime in the Arpa River Basin. The fitted semivariogram models for summer show that the Exponential fitted model has the longest range of approximately 31.9 km, indicating greater spatial continuity of groundwater than the Spherical fitted model which has a range of approximately 24.4 km. The Spherical fitting to a semivariogram indicates a broader range for the fitting than the Circular fitted model, which shows a moderate range of 23.3 km. The Gaussian model has the shortest fitting range at approximately 18.4 km, indicating a stronger smoothing effect with lesser accuracy over shorter distances.

The effectiveness of all models was evaluated via a variety of cross-validation statistics such as Mean Error (ME), Root Mean Square Error (RMSE), Average Standard Error (ASE), Mean Standardized Error (MSE), and Root Mean Square Standardized Error (RMSSE). The model favorability was determined based on the integrated information of all the mentioned statistics rather than on one specific indicator. Lower RMSE values were prioritized due to their representation of the accuracy of the predictions while values of ME and MSE close to zero indicate lack of bias of predictions. Moreover, similar ASE, RMSE, and RMSSE values close to one proved to be a good indication of reliable uncertainty estimation. Therefore, the best model per season was selected by looking at the results of all validation statistics.

The visual interpretation of the interpolation maps was supported by the cross-validation statistics presented in Tables 2, 3 and 4. Models producing lower RMSE values, ME values close to zero, and better agreement between RMSE and ASE were considered to provide more reliable spatial predictions. Therefore, the interpretation of the interpolation maps was based on both the visual characteristics of the generated groundwater surfaces and the quantitative validation results rather than on visual assessment alone. The Spherical semivariogram produced comparatively smoother and spatially consistent groundwater level surfaces while also exhibiting favourable cross-validation statistics. Therefore, its selection was supported by both quantitative validation results and visual interpretation of the interpolation maps. Compared with Ordinary Kriging, the IDW interpolation produced relatively localized variations around observation wells, whereas the Spline method generated comparatively smoother surfaces. These differences are

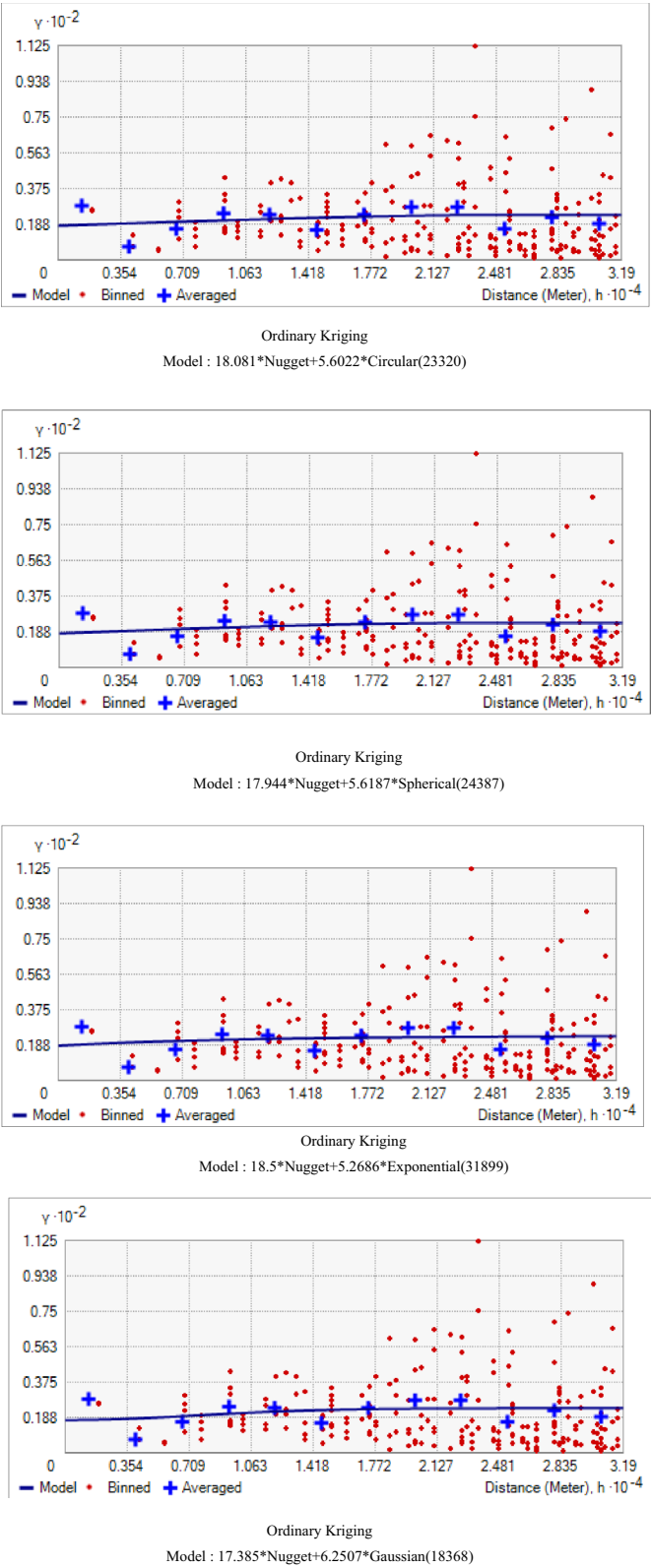


Fig. 5 Experimental Semivariogram and fitted circular model for average groundwater table (mbgl) during the summer season

Table 2 Geostatistical analysis on rainy season average water table

Parameter	Ideal	OK Circular	OK Spherical	OK Exponential	OK Gaussian	IDW	Spline
Mean Error (ME)	≈ 0	0.0507	0.032	0.0282	0.060262	−0.080	–
RMSE	Minimum	3.8419	3.8013	3.787	3.893661	0.338	–
Mean Standardized	≈ 0	0.0109	0.0063	0.0051	0.014792	–	–
RMS Standardized	≈ 1	1.0217	0.9953	0.9426	1.077268	–	–
ASE	≈ RMSE	3.8446	3.8889	4.0795	3.664683	–	–

Table 3 Geostatistical analysis on winter season average water table

Parameter	Ideal	OK Circular	OK Spherical	OK Exponential	OK Gaussian	IDW	Spline
Mean Error (ME)	≈ 0	0.0572	0.0392	0.0344	0.063097126	−0.084	–
RMSE	Minimum	4.0487	4.0085	3.9939	4.103969234	0.442	–
Mean Standardized	≈ 0	0.0117	0.0075	0.0062	0.015239405	–	–
RMS Standardized	≈ 1	0.9968	0.9725	0.9167	1.056032139	–	–
ASE	≈ RMSE	4.1525	4.1981	4.4187	3.94610098	–	–

Table 4 Geostatistical analysis on summer season average water table

Parameter	Ideal	OK Circular	OK Spherical	OK Exponential	OK Gaussian	IDW	Spline
Mean Error (ME)	≈ 0	0.0731	0.0738	0.0486	0.04987971	−0.073	–
RMSE	Minimum	4.3168	4.3325	4.3308	4.366207974	0.304	–
Mean Standardized	≈ 0	0.0141	0.0142	0.0087	0.010170699	–	–
RMS Standardized	≈ 1	0.933	0.9341	0.92	0.951915635	–	–
ASE	≈ RMSE	4.6953	4.7064	4.7686	4.650727955	–	–

consistent with the characteristics of deterministic interpolation methods and were evaluated together with the cross-validation statistics reported in Tables 2, 3 and 4.

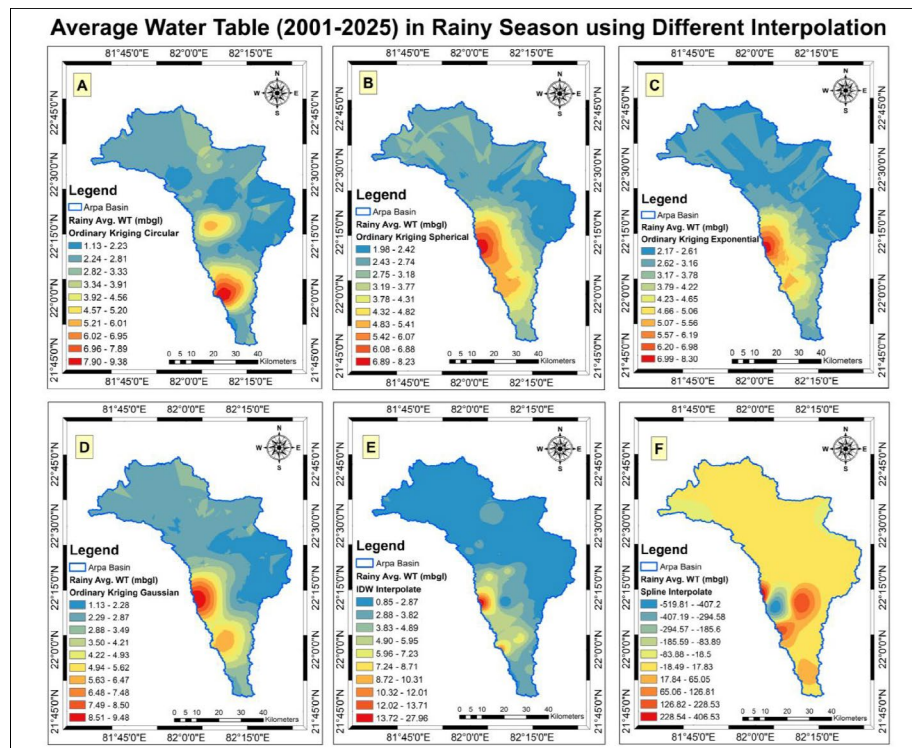
The Geostatistical Analysis is done on above Interpolation techniques shows in Table 2 of Rainy Season Groundwater Table (mbgl) data to assess the suitable method for spatial representation. During the rainy season, the Exponential semivariogram produced the lowest RMSE (3.787 m), the smallest Mean Error (0.0282), and the Mean Standardized Error closest to zero, indicating comparatively higher prediction accuracy. However, the Spherical model yielded an RMS Standardized Error (0.9953) closest to the ideal value of one and showed good agreement between ASE and RMSE, indicating more reliable uncertainty estimation. Therefore, considering all cross-validation statistics together, the Spherical model demonstrated balanced overall performance, whereas the Exponential model showed slightly better prediction accuracy.

The Geostatistical Analysis is done on above Interpolation techniques shows in Table 3 of Winter Season Groundwater Table data to assess the suitable method for spatial representation. During the winter season, the Exponential semivariogram model produced the lowest RMSE (3.9939 m), the smallest Mean Error (0.0344), and the Mean Standardized Error closest to zero, indicating comparatively higher prediction accuracy. In contrast, the Circular semivariogram yielded an RMS Standardized Error (0.9968) closest to the ideal value of one and demonstrated close agreement between ASE (4.1525) and RMSE (4.0487), indicating more reliable uncertainty estimation. As no single semivariogram model performed best for all cross-validation statistics, model selection was based on the combined interpretation of prediction accuracy (ME and RMSE) and uncertainty measures (ASE and RMSSE), rather than on a single statistical criterion. Accordingly, although the Exponential model provided marginally higher prediction accuracy, the

Circular model demonstrated the most balanced overall performance for winter season groundwater level interpolation.

Geostatistical analysis was carried out for summer season groundwater table data to assess the performance of various interpolation techniques shows in Table 4 for spatial representation and further research use in diverse field. During the summer season, the Exponential semivariogram model produced the smallest Mean Error (0.0486), the lowest Mean Standardized Error (0.0087), and an RMSE (4.3308 m) comparable to the other semivariogram models, indicating relatively higher prediction accuracy. In contrast, the Gaussian semivariogram yielded an RMS Standardized Error (0.9519) closest to the ideal value of one and showed better agreement between ASE (4.6507) and RMSE (4.3662), indicating comparatively reliable uncertainty estimation. Since no single semivariogram model consistently outperformed the others for all validation statistics, model selection was based on the combined interpretation of prediction accuracy (ME and RMSE) and uncertainty measures (ASE and RMSSE). Accordingly, the Exponential model demonstrated comparatively higher prediction accuracy, whereas the Gaussian model exhibited more balanced uncertainty estimation during the summer season.

Different methods of interpolation have shown in Fig. 6 significantly different spatial distributions of groundwater tables during the rainy season. Of all the methods used, Ordinary Kriging using the Spherical Semivariogram model produces the most realistic and coherent contour of the groundwater surface, with smooth transitions and clearly defined high and low groundwater zones. This is supported by the lowest RMSE, and



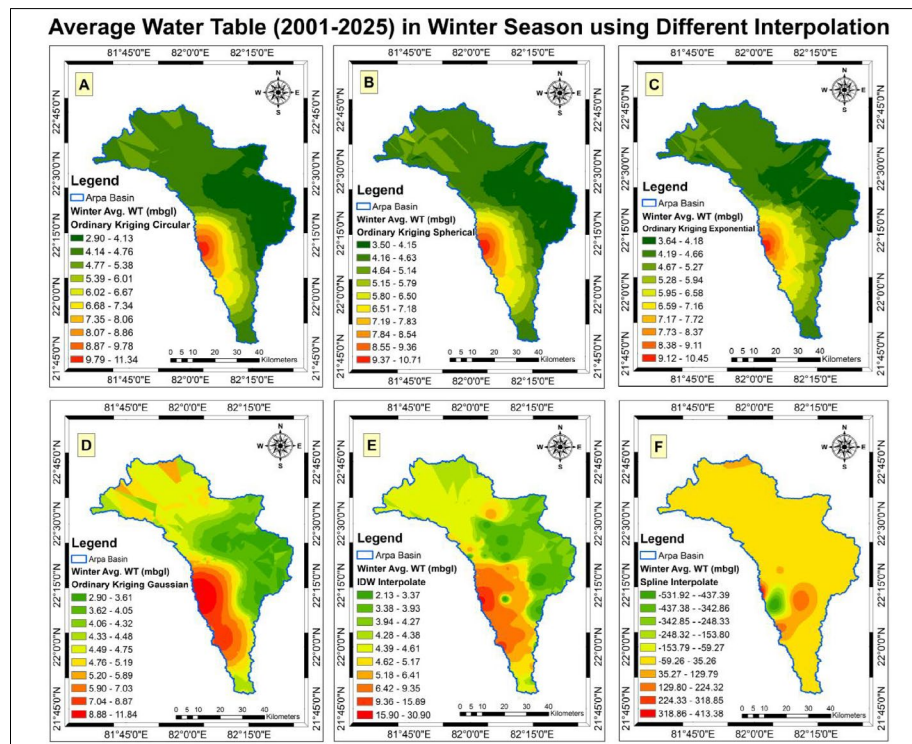
Ordinary Kriging (A-Circular, B- Spherical, C- Exponential, D- Gaussian)
Deterministic Interpolation (E- IDW and F- Spline)

Fig. 6 Spatial representation of rainy season groundwater table(mbgl) in Arpa River Basin

a ME close to 0 obtained from geostatistical validation. In comparison, the IDW and Spline methods exhibit localized over-exaggeration and/or over-smoothing, leading to greater uncertainty, whereas the kriging-based contour maps exhibit better accuracy, reliability, and agree in both visual interpretation and statistical performance.

Spatial boundaries of water table depths in the Arpa River Basin are consistent across each of the interpolation maps in Fig. 7 that depict groundwater levels during winter. Water table depths in the southern and central portions of the basin contain the higher concentrations of water table depth. The most effective method for producing water table depth surfaces that were both spatially consistent and hydrogeologically realistic was the Spherical Semivariogram model of Ordinary Kriging, which produced water table depth surfaces that adequately captured gradual transitions and localized variations, and were consistent with geostatistical validation results. The IDW and Spline methods produced exaggerated extremes as well as high uncertainty due to lack of spatial continuity. The water table depth surfaces produced by kriging provided the best representation of winter-season groundwater conditions.

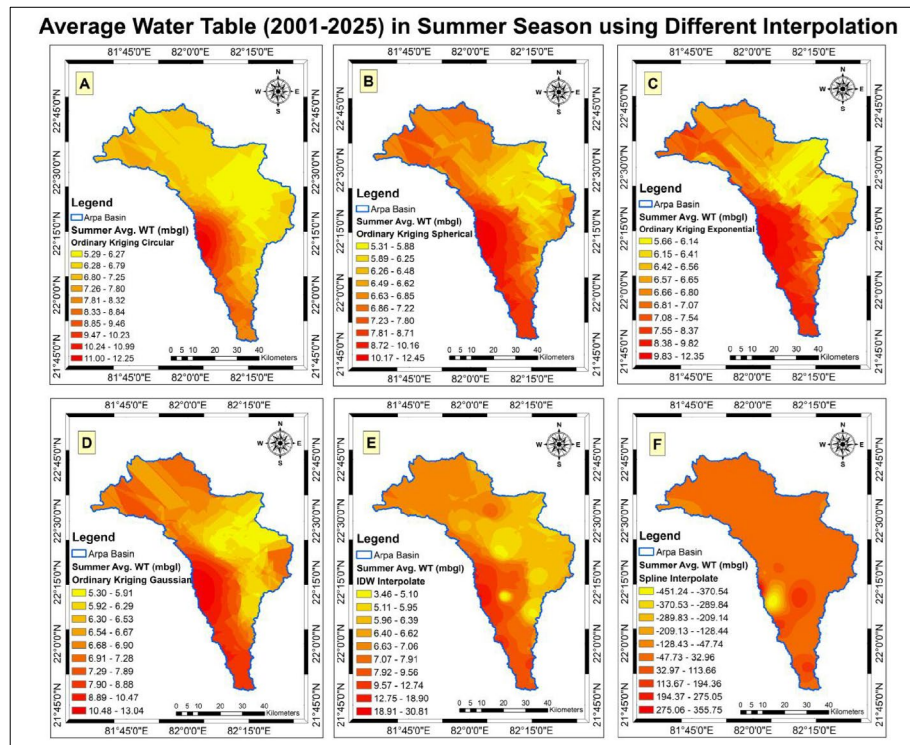
Interpolation maps depicting summer season groundwater table depths indicate in Fig. 8a large amount of spatial variation with the case being more evident in the southern and central portions of the Arpa River Basin. Drought-year groundwater table depths are deeper than other areas that would indicate that there is a significant amount of groundwater stress during the dry months of the year in these locations as well. From the various testing techniques evaluated, Ordinary Kriging using a Spherical Semivariogram model created the most representative and spatially balanced representation of



Ordinary Kriging (A-Circular, B- Spherical, C- Exponential, D- Gaussian)

Deterministic Interpolation (E- IDW and F- Spline)

Fig. 7 Spatial representation of winter season groundwater table(mbgl) in Arpa River Basin



Ordinary Kriging (A-Circular, B- Spherical, C- Exponential, D- Gaussian)

Deterministic Interpolation (E- IDW and F- Spline)

Fig. 8 Spatial representation of summer season groundwater table(mbgl) in Arpa River Basin

the groundwater surface as it clearly captured the regional trends as well as localized variability, and provided the most consistent result using the geostatistical performance measures. The IDW and Spline methods have been shown to create local exaggerations and smoothing that may not accurately reflect real-world conditions and thus produce biased results for summer groundwater distribution. On the contrary, kriging-generated outputs provide the best possible representation of how summer groundwater is geographically distributed.

The different interpolation modes showed different readings in validation statics, selection of the model was based on complete validation statistics, which used the cross-validation at the same time. Results of the experiments with Spherical semivariogram showed low RMSE values and ME close to zero with good agreement of obtained RMSE, ASE and RMSSE values during three seasons. In conclusion, consulting the RMSE statistics based on Spherical semivariogram demonstrated the best results in prediction of seasonal groundwater level in the Arpa River Basin.

4 Conclusion

The current research showed that the efficiency of different spatial interpolation techniques for estimating the level of groundwater changes according to climatic conditions and this highlights the necessity of analyzing the results obtained using different models of interpolation separately for each of the seasons of the year. Different deterministic (IDW and Spline) and geostatistical (Ordinary Kriging employing Circular, Spherical,

Exponential, and Gaussian semivariogram models) methods of interpolation were compared using a database of groundwater level observations for an extended period (2001–2025) from Arpa River Basin.

Analysis of cross-validation results showed that there was no semivariogram model that performed better than other models for any of the validation statistics. The Exponential semivariogram performed better in terms of prediction error as can be seen from low values of the Mean Error (ME), Root Mean Square Error (RMSE), and Mean Standardized Error for the three seasons, thus indicating the high predictive value of the model. However, when it comes to uncertainty statistics all the three models in the study (i.e. Circular, Spherical and Gaussian semivariogram) behaved differently depending on the seasons. Hence, model selection should be based on several different statistics.

In summary, findings show that geostatistical interpolation techniques yield better and spatially consistent groundwater level estimations compared to deterministic methods for mapping seasonal groundwater levels for the Arpa River Basin. These seasonal groundwater level maps can serve as important sources of spatial information for groundwater evaluation, watershed management, groundwater conservation planning, and sustainable use of water resources. Future studies may further enhance interpolation precision through the disposal of more hydrological variables, denser monitoring networks, and advanced geostatistical or machine learning techniques.

Abbreviations

CGWB	Central Ground Water Board
OK	Ordinary Kriging
IDW	Inverse Distance Weighted
RMSE	Root Mean Square Error
ME	Mean Error
ASE	Average Standard Error
RMSSE	Root Mean Square Standardized Error
mbgl	Meter below ground level

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Author contributions

The corresponding author Khirsagar Patel contributed to the conceptualization of the study and writing of the original draft. The other two authors Dr. Prasoon Soni and Prof. Pushpraj Singh contributed to supervision, validation of results, and overall review of the manuscript.

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Data availability

Data available on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for application

Not applicable.

Competing interests

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